Applied NLP

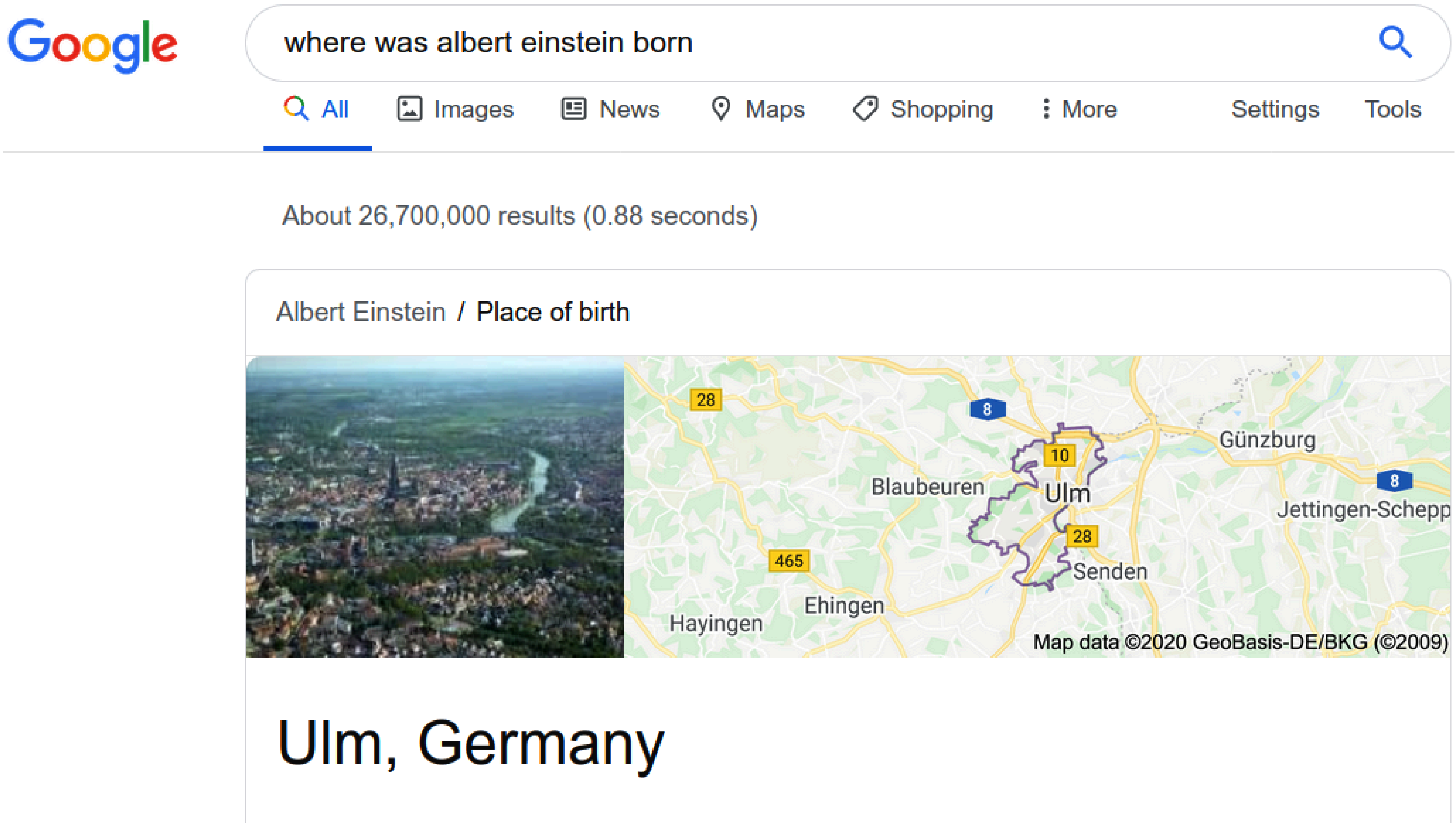
DATA 641

Lecture 11 — Named Entity Recognition

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

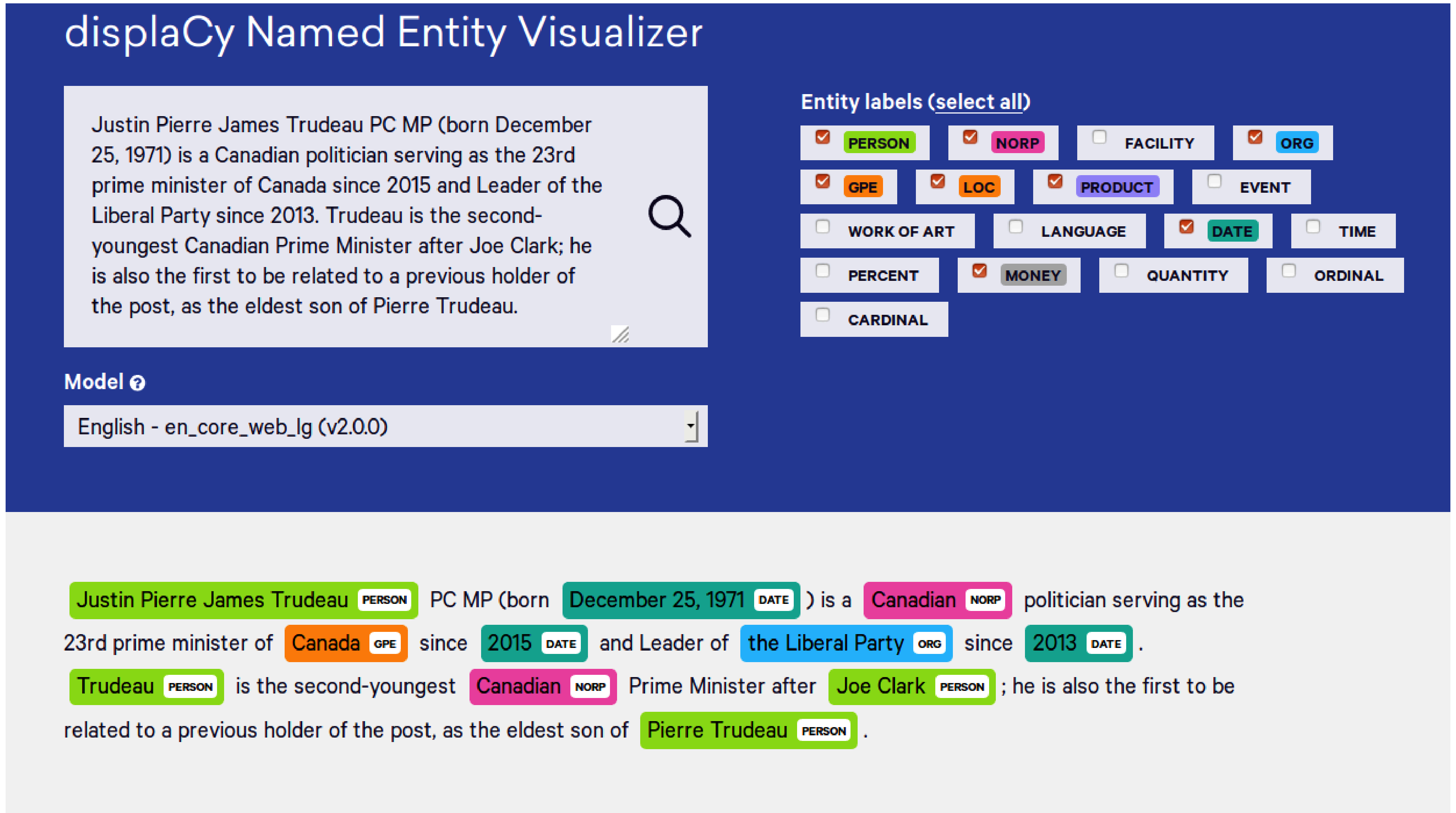
Named Entity Recognition (NER)

- Consider a scenario where the user asks a search query—"Where was Albert Einstein born?"—using Google search. To be able to show "Ulm, Germany" for this query, the search engine needs to decipher that Albert Einstein is a person before going on to look for a place of birth. This is an example of NER in action in a real-world application.



- NER refers to the IE task of identifying the entities in a document. Entities are typically names of persons, locations, and organizations, and other specialized

- NER is an important step in the pipeline of several NLP applications involving information extraction. The following figure illustrates the function of NER using the `displaCy` visualizer by explosion.ai



- As seen in the figure, for a given text, NER is expected to identify person names, locations, dates, and other entities. Different categories of entities identified here are some of the ones commonly used in NER system development.

Note 1: NER is a prerequisite for being able to do other IE tasks, such as relation extraction or event extraction, which were introduced earlier in this chapter and will be discussed in greater detail later on.

Note 2: NER is also useful in other applications like machine translation, as names need not necessarily be translated while translating a sentence. So, clearly, there is a range of scenarios in NLP projects where NER is a major component. It is one of the common tasks you are likely to encounter in NLP projects in industry.

Building an NER System

- A simple approach to building an NER system is to maintain a large collection of person/ organization/location names that are the most relevant to our company (e.g., names of all clients, cities in their addresses, etc.); this is typically referred to as a gazetteer.

- To check whether a given word is a named entity or not, just do a lookup in the gazetteer. If a large number of entities present in our data are covered by a gazetteer, then it is a great way to start, especially when we do not have an existing NER system available.

- How does it deal with new names? How do we periodically update this database? How does it keep track of aliases, i.e., different variations of a given name (e.g., USA, United States, etc.)?

- An approach that goes beyond a lookup table is rule-based NER, which can be based on a compiled list of patterns based on word tokens and POS tags. For example, a pattern "NNP was born," where "NNP" is the POS tag for a proper noun, indicates that the word that was tagged "NNP" refers to a person.

- A more practical approach to NER is to train an ML model, which can predict the named entities in unseen text. For each word, a decision has to be made whether or not that word is an entity, and if it is, what type of the entity it is. In many ways, this is very similar to the classification problems. The only difference here is that NER is a "sequence labeling" problem. The typical classifiers predict labels for texts independent of their surrounding context.

    - Consider a classifier that classifies sentences in a movie review into positive/negative/neutral categories based on their sentiment. This classifier does not (usually) take into account the sentiment of previous (or subsequent) sentences when classifying the current sentence.

    - In a sequence classifier, such context is important. A common use case for sequence labeling is POS tagging, where we need information about the parts of speech of surrounding words to estimate the part of speech of the current word.

    - NER is traditionally modeled as a sequence classification problem, where the entity prediction for the current word also depends on the context. For example, if the previous word was a person name, there is a higher probability that the current word is also a person name if it is a noun (e.g., first and last names).